Exercise Sheet 1

Uncertainty, Decisions and Reasoning

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer



1.A Decisions From Probabilities

Exercise 1.A.1: Three Loan Applicants

A loan model gives three applicants a **default probability** of 0.05, 0.35 and 0.45. Accepting an applicant who defaults costs the bank **€8,000**; rejecting one who would have repaid costs **€500** in lost margin.

1. Which applicants does the rule "reject if $p > 0.5$" reject?
2. Compute the expected cost of *accepting* and of *rejecting* each applicant. Which decision minimises it?
3. Derive the break-even probability p^* in general, for a cost c_A of accepting a defaulter and c_R of rejecting a good customer. Evaluate it here.
4. How does p^* change if the bank's margin doubles? If the loan is ten times larger (and the margin stays)? Explain the direction in one sentence each.

Exercise 1.A.2: Allow, Block — or Ask a Human?

A fraud model scores a transaction with probability p of fraud. Three actions:

- **allow**: if it is fraud, the bank loses **€900**;
- **block**: if it is legitimate, an annoyed customer costs **€300**;
- **review**: an analyst checks it for a flat **€60** and always decides correctly.

1. Write the expected cost of each action as a function of p.
2. For which range of p is each action optimal?
3. Without a review option, where would the threshold between allow and block be? What does the review option change?

Exercise 1.A.3: Choosing a Threshold From Data (Python)

```python
import numpy as np
rng = np.random.default_rng(1)
p = rng.beta(0.5, 6, size=10_000)        # model scores (assume calibrated)
y = rng.random(10_000) < p               # true labels drawn from the scores
c_A, c_R = 8_000, 500                    # costs as in 1.A.1
```

1. For thresholds $t \in \{0, 0.01, \ldots, 1\}$ ("reject if $p > t$"), compute the **average realised cost** per applicant.
2. Which t minimises it? Compare with p^* from 1.A.1.
3. What would happen to the best t if the scores were systematically too low (say `p_reported = p / 2`)?

1.B Where Uncertainty Comes From

Exercise 1.B.1: Noise or Ignorance?

For each case, say whether the uncertainty is mainly **aleatoric** (in the problem), **epistemic** (in our knowledge), or both — and whether more data of the same kind would help.

1. Predicting the outcome of a fair coin flip.
2. A house-price model trained on flats of 40–120 m^2, asked about a 600 m^2 villa.
3. A spam filter trained on English mail, applied to Finnish mail.
4. Two customers with identical records; one churns, one does not.
5. A clinical measurement with known device noise, taken once per patient.

Exercise 1.B.2: One Pipeline, Many Uncertainties

A bank trains a logistic-regression fraud model on last year's transactions. Labels come from chargebacks, which arrive up to 90 days after the transaction. The test set holds 2,000 recent transactions, of which **3** are fraud.

Name **five** distinct places where uncertainty enters this system. For each, say which box of the learning system it lives in and propose one mitigation.

1.C Reasoning Under Uncertainty

Exercise 1.C.1: Which Mode of Reasoning?

Classify each step as **deduction**, **induction** or **abduction**.

1. "All training examples with a foreign IP and a night-time purchase were fraud, so such transactions are fraud."
2. "The model's accuracy dropped last week; the most plausible explanation is the new payment provider."
3. "If the logistic model is correct, doubling feature x_1 multiplies the odds by $e^{w_1 x_1}$."
4. "Customers who use the app churn less; perhaps loyal customers install the app."
5. "Gradient descent with this learning rate on this convex loss converges."

Exercise 1.C.2: Save the Model or Change It?

A fraud model has been reliable for two years. After a new product launch, its recall on the new product's transactions falls from 0.8 to 0.4.

1. Name the "save the theory" and the "change the theory" move in this setting.
2. Design two cheap checks, in order, that would tell the moves apart.
3. What evidence would make you abandon the current model family — and why should one bad week not be enough?